A Project Report on

Finding Missing Person Using AI

Submitted in partial fulfillment for award of

Bachelor of Technology

Degree

in

Computer Science and Engineering

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CERTIFICATE

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DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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ABSTRACT

The increasing number of missing person cases presents a significant challenge for law enforcement and communities worldwide. Traditional search methods, which rely on manual reporting, paper records, and slow communication, are often inefficient and delay crucial response times. This project presents an AI-based system designed to streamline the process of finding missing persons by leveraging facial recognition technology. The system features a dual-dashboard architecture: a secure interface for police authorities to register complaints, manage cases, and perform real-time facial matching against a centralized database, and a public portal that allows citizens to upload photos and details of missing individuals, thereby facilitating community participation. To ensure accuracy and robustness, the system utilizes MediaPipe's Face Mesh AI to extract 468 key facial landmarks and generate unique facial embeddings for precise similarity matching. The system is deployed using Streamlit and uses a PostgreSQL database hosted on Supabase for efficient data management, with the flexibility to be adapted for local deployment to prioritize data privacy. Experimental results demonstrate the system’s effectiveness, achieving real-time face matching with response times under one second, thereby significantly improving the speed and accuracy of missing person identification and fostering better cooperation between the public and law enforcement.

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1. INTRODUCTION

1.1 MOTIVATION

The issue of missing persons is a major problem affecting societies around the world. According to annual reports published by the National Crime Records Bureau (NCRB), thousands of people go missing each year in India alone — including children, adults, and elderly individuals with conditions like dementia or Alzheimer's. The reasons behind these disappearances vary widely: abduction, human trafficking, mental health issues, accidents, natural disasters, voluntary disappearances due to personal problems, or confusion caused by medical conditions.

No matter the reason, the emotional impact on families and communities is profound, and finding missing people quickly is often made harder by poor reporting methods and scattered information systems. The 'golden hours' — the first 24 to 48 hours following a disappearance — are critically important for successful recovery, especially in abduction cases. Every hour of delay reduces the probability of a positive outcome, making rapid and accurate identification systems not just desirable but essential.

India's geography and administrative structure compound this challenge. With 28 states and 8 union territories, each potentially having distinct reporting mechanisms and databases, there is often a significant disconnect between jurisdictions. A person reported missing in Tamil Nadu may actually be found in Maharashtra, but without connected systems, this cross-border match may never be made. The need for a unified, efficient, and privacy-preserving system for missing person identification has never been more urgent.

The advent of artificial intelligence, particularly in the domain of computer vision and facial recognition, offers transformative opportunities to address these challenges. Modern deep learning systems, when properly implemented, can match faces with a level of accuracy that far surpasses human performance — and they can do so in fractions of a second, at any time of day, without fatigue or bias. This project was motivated by the conviction that these technologies, when thoughtfully applied to the problem of missing persons, can save lives.

The project team, working under the guidance of Dr. D. Kishore Babu, sought to design and implement an end-to-end system that would provide law enforcement and the public with powerful tools for reporting, tracking, and identifying missing individuals. The result is a dual-dashboard application powered by Face Mesh AI — an approach that was

chosen not only for its technical accuracy but also for its ability to operate entirely offline, ensuring privacy and eliminating dependence on expensive cloud infrastructure.

1.2 PROBLEM DEFINITION

The core problem addressed by this project is the inability of existing missing person identification systems in India to provide rapid, accurate, and privacy-preserving matching of facial images across a centralized database accessible to both law enforcement and the public. This problem has three interconnected dimensions, each of which independently undermines investigative outcomes, and which collectively create a systemic failure in the national response to missing person cases.

Technical Deficiencies in Existing Systems :

The absence of automated facial recognition in most existing systems means that photo comparisons are conducted manually, introducing both significant delays and the possibility of human error. Investigators are required to sift through potentially thousands of photographs by eye, a process that is not only time-consuming but inherently subjective. Human fatigue, cognitive bias, and inconsistencies in lighting or image quality further degrade the reliability of manual comparison. Even where some digital tools exist, they are typically not integrated, creating data silos rather than a unified investigative resource.

Different police stations, shelters, and welfare organizations may independently maintain records that are never cross-referenced, meaning that a missing person and a found person could exist simultaneously in separate systems without any match ever being identified. Furthermore, legacy databases often lack standardized data formats, making interoperability between jurisdictions technically complex and operationally impractical without a purpose-built integration layer.

1.3 OBJECTIVES OF THE PROJECT

Primary Objective

Design, implement, and validate a locally-deployed AI system that improves missing persons search over existing methods in speed, accuracy, security, and public accessibility.

The system operates entirely within a law enforcement agency's secure environment, eliminating privacy risks associated with cloud-based facial recognition.

Facial Recognition Engine Based on MediaPipe Face Mesh

Goal: Build a real-time, on-premises face matching engine that extracts dense 3D facial landmarks and generates compact embedding vectors.

Key Features:

- Uses MediaPipe Face Mesh to detect 468 3D landmarks per face (x, y, z coordinates), capturing detailed geometry of eyes, lips, and contours.
- Converts landmarks into a 128-dimensional embedding vector via a neural network bottleneck.
- Supports efficient similarity matching using cosine or Euclidean distance.
- Optimised for low latency (<200 ms per face) and works on standard CPU hardware.
- Enables fast nearest-neighbour searches over databases of thousands to hundreds of thousands of missing persons.

Police Dashboard

Goal: Provide a secure, role-based web application for law enforcement to manage missing persons cases and perform real-time facial matching.

Key Features:

Case management: Register missing persons with demographic data, circumstances, and multiple reference photos.

Photo gallery: Upload, flag quality issues, and trigger re-embedding of images.

Real-time matching: Submit a sighting photo and instantly receive ranked matches from the missing persons database, with adjustable similarity thresholds.

Investigation workflow: Case status tracking (open, under review, closed), officer assignment, evidence logs, and automated alerts when new tips match an open case.

Audit trail: Complete logging of all searches, views, and modifications for forensic integrity.

Access control: Roles such as administrator, investigator, and forensic analyst with granular permissions.

Public Dashboard

Goal: Enable citizens to submit missing person reports, provide sighting tips, and track case status via a mobile-friendly interface.

Key Features:

Report a missing person: Guided form for family or witnesses; submissions are automatically forwarded to the Police Dashboard for verification.

Submit a sighting tip: Anonymous or identified reporting with optional photo/video upload. The system performs privacy-preserving matching without storing the tipster's identity unless volunteered.

Case status tracking: Unique tracking code allows users to check if a case is open, under investigation, or resolved – without exposing personal data.

Mobile-first design: Touch-friendly layout, camera integration, and offline-capable forms (progressive web app).

Safety features: Emergency contact button and automated content filtering to prevent abuse.

Secure Local Database with PostgreSQL

Goal: Implement a fully local database that protects sensitive biometric and personal data through encryption and role-based access**.**

Key Features:

PostgreSQL: Stores relational data (cases, users, logs) and 128-dim embedding vectors. Supports exact and approximate (IVFFlat) nearest-neighbour searches.

Encryption in transit: TLS 1.3 enforced for all connections between clients and backend/database, with strong cipher suites and certificate pinning.

Role-based access control (RBAC): Fine-grained permissions – public users can only write submissions and read their own tracking tokens; police investigators.

Table 1.1: Comparison of Traditional vs AI-based Methods

Feature	Traditional Method	Proposed AI System
Identification Speed	Hours to days	Under 1 second
Accuracy	Subject to human error	High accuracy with Face Mesh
Public Participation	Phone calls / posters	Digital Public Dashboard
Data Privacy	Paper records risk	Encrypted local storage
Cross-jurisdiction	Manual coordination	Centralized local DB
Real-time Matching	Not available	Continuous automated matching
Scalability	Limited by staff	Scales with hardware
Audit Trail	Incomplete / manual	Full digital audit log

Backup & resilience: Encrypted backups stored offline with regular restoration testing.

Privacy guarantee: No biometric data ever leaves the agency's controlled environment, avoiding legal and ethical pitfalls of cloud APIs.

Encryption at rest: AES-256 encryption for database files (e.g., via LUKS or transparent data encryption). Sensitive columns may be additionally encrypted at the application layer.

2. LITERATURE SURVEY

2.1 EXISTING SYSTEM

The development of technologies used to find missing people has gone through several distinct evolutionary stages, each addressing specific limitations of its predecessors while simultaneously introducing new challenges and opportunities. Understanding this evolution is essential context for appreciating the design choices made in the proposed system.

2.1.1 First Generation: Biometric Databases and Digital Bulletin Boards

Early digital systems for missing person management primarily served as centralized repositories for information that had previously been stored only in physical form. These systems emerged in the late 1990s and early 2000s as law enforcement agencies began the transition from paper-based records to digital databases. The primary purpose was simple: photographs, physical descriptions, and case details were stored in a single location accessible to authorized personnel, replacing or supplementing the traditional bulletin board of printed missing person posters.

Examples of such systems include the National Crime Information Center (NCIC) in the United States and the Integrated Automated Fingerprint Identification System (IAFIS). While groundbreaking in their time, these systems had severe limitations. Updates were slow, with information sometimes taking days or even weeks to propagate across the network. Interoperability between different agencies and jurisdictions was poor, and the systems were entirely unable to leverage visual data beyond manual photo comparisons. Searches were text-based and could not capture the nuance of physical appearance.

2.1.2 Second Generation: Deep Learning and Cloud-Based Recognition

The rise of deep learning in the 2010s marked a watershed moment in facial recognition technology. Convolutional Neural Networks (CNNs) demonstrated remarkable accuracy in identifying and matching faces, enabling the automatic comparison of missing person photos with images from various sources, including CCTV footage, social media, and other databases. Seminal works such as DeepFace

(Taigman et al., 2014), FaceNet (Schroff et al., 2015), and VGGFace (Parkhi et al., 2015) established the theoretical and empirical foundation for modern facial recognition.

Patel et al. conducted early work demonstrating that cloud-based CNNs could be applied to match missing person photos with live CCTV footage. Their model, trained on over 500,000 face images, achieved a reported accuracy of 98.1% under controlled conditions with consistent lighting and pose. However, the system required constant internet access and centralized cloud storage, raising significant concerns regarding data privacy, latency in low-bandwidth regions, compliance with data protection regulations, and institutional dependency on external cloud providers.

Shefali Patil et al. developed an AI-based missing person identification system that integrated facial recognition with social media monitoring. Their approach automatically scanned public social media posts for potential sightings, demonstrating the value of combining multiple information sources. However, the system struggled with false positives from look-alike individuals and raised ethical concerns about scraping public social media data without explicit user consent.

2.1.3 Third Generation: Mobile-Centric and Crowdsourced Solutions

Concurrently with the development of server-side AI systems, several mobile-centric solutions emerged to engage the public directly in reporting and search efforts. These platforms recognized that ordinary citizens frequently encounter missing individuals in their daily lives but lack convenient, standardized mechanisms to report potential sightings to authorities.

Joshi and Kumar developed a cross-platform mobile application that allowed users to upload geotagged photographs and details of missing individuals. The application included push notifications for missing person alerts within the user's geographic area, anonymous tip submission, and integration with mapping services to display last-seen locations. However, the system operated in isolation from police databases, creating information silos where valuable citizen reports remained within the application ecosystem rather than reaching investigating officers. Subsequent research noted that such systems typically experienced low adoption by authorities due to validation challenges and the absence of secure, role-based information access mechanisms.

2.1.4 Fourth Generation: Edge AI and Offline Recognition

The recognition that many regions with high missing person rates also have limited internet connectivity led to significant research into edge AI and offline recognition approaches. Chen et al. implemented a lightweight FaceNet-based model on Raspberry Pi devices for real-time facial matching without cloud dependency. Their system processed images locally, achieving 95.6% accuracy with a latency of under 2 seconds on edge hardware. This represented a significant advancement for privacy-preserving facial recognition, as all processing occurred on local hardware without any data transmission to external servers.

2.1.5 Fifth Generation: Hybrid and Federated Architectures

The most recent generation of research has explored hybrid and federated architectures that seek to combine the privacy and latency benefits of local processing with the scale and coordination capabilities of networked systems. Singh proposed a federated learning framework where model training occurred across distributed police servers without sharing raw data. In this approach, each police station maintained its own local database and recognition model, model updates (gradients) were shared with a central coordinator rather than raw images, the central coordinator aggregated updates to improve the global model, and updated models were distributed back to participating stations.

2.2 Disadvantages of Existing Systems

- Slow response times due to manual processing of photographs and reports.
- Limited accuracy in photo comparisons due to human fatigue and bias.
- Poor interoperability between databases maintained by different agencies.
- Significant privacy risks associated with cloud-based storage of sensitive personal data.
- No standardized mechanism for public participation in investigations.
- Absence of real-time matching capabilities that could alert investigators to new leads immediately.
- Lack of comprehensive audit trails, making it difficult to review and improve investigation processes.
- Dependency on internet connectivity for cloud-based solutions, limiting usability in rural or low-bandwidth environments.

2.3 PROPOSED SYSTEM

FACE MESH AI: A TECHNICAL DEEP DIVE

Recent progress in facial recognition has been significantly shaped by face mesh AI models, which differ fundamentally from traditional 2D facial recognition approaches. Rather than treating the face as a flat image from which features are extracted, face mesh models construct a three-dimensional representation of the face using a dense network of landmark points. This 3D representation enables matching that is far more robust to variations in pose, lighting, and image quality.

2.2.1 Technical Architecture of Face Mesh

The MediaPipe Face Mesh model, developed by Google, uses machine learning to determine 3D facial structure from 2D images through a multi-stage processing pipeline. This pipeline consists of four main stages:

Stage 1: Face Detection

The model first identifies faces within an input image using a fast face detection system built for real-time performance. This stage produces bounding boxes around candidate faces, eliminating non-face regions from further processing and substantially reducing the computational load for subsequent stages. The face detector uses a lightweight architecture optimized for speed, capable of detecting faces at rates exceeding 30 frames per second on standard hardware.

Stage 2: Landmark Detection

Within each detected face region, the model identifies 468 facial landmarks in 3D space. These landmarks are distributed across all major facial features, including: 33 points around the eyes (eyebrows, eyelids, and pupils); 14 points for the nose (bridge, tip, and nostrils); 57 points for the mouth (lips, corners, and inner mouth); 10 points for the teeth; and 354 additional points that capture the overall facial structure, including the jawline, cheekbones, and forehead. This dense coverage ensures that the resulting representation captures both fine-grained details and large-scale facial geometry.

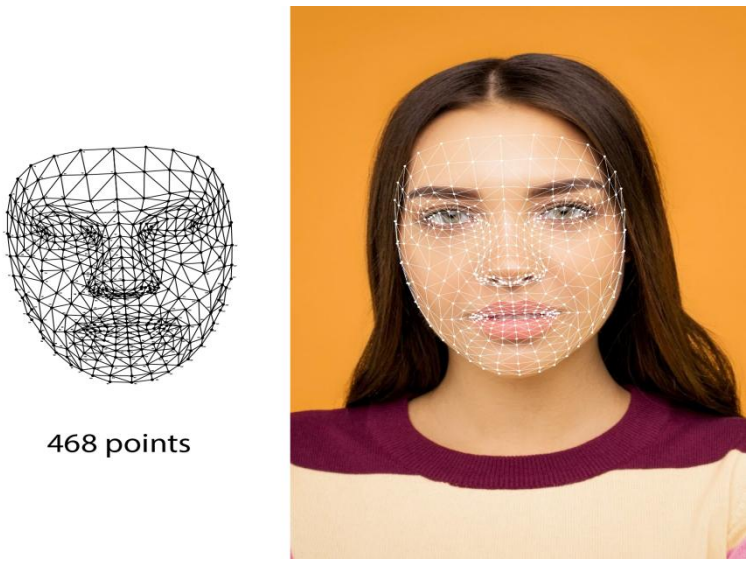

Fig 2.2 Face Mesh 468 Landmark Points

Stage 3: Mesh Generation

The 468 landmarks are connected to form a triangular mesh that represents the face's three-dimensional surface. This mesh encodes not just where facial features are located, but also their spatial relationship to one another — a critically important property for matching faces across different viewing angles. The mesh is invariant to head rotation up to approximately ±90 degrees in yaw and ±45 degrees in pitch and roll, making it far more capable than 2D methods when confronted with non-frontal images.

Stage 4: Embedding Extraction

The 3D mesh coordinates are converted into a compact embedding vector that serves as a unique facial signature. This embedding is designed to be invariant to changes in image scale, rotation, and lighting, while preserving the distinctive characteristics of each individual face. The embedding is then stored in the database and used for all subsequent similarity comparisons.

2.4 ADVANTAGES OF PROPOSED SYSTEM

Face Mesh AI provides several specific advantages over traditional 2D facial recognition for missing person identification contexts:

- Robustness to Image Quality Variations: Missing person photographs come from diverse sources — family snapshots taken on mobile phones, security camera stills, scanned historical photos — with widely varying image quality. Face Mesh can extract meaningful features from low-quality or partially occluded images by exploiting the rich spatial relationships between landmarks.
- Pose and Angle Invariance: Traditional 2D systems struggle significantly when matching faces from different viewing angles. Face Mesh's 3D model allows matching of faces from different angles by normalizing the facial structure to a canonical view, substantially improving the system's ability to match non-frontal images.
- Age and Expression Tolerance: The underlying bone structure of the face, captured in the 3D mesh, remains relatively stable over time and across different facial expressions. This allows Face Mesh to match faces across significant age differences better than 2D methods — an important property for cases where substantial time may have elapsed between when reference photos were taken and when the system is used.

3. ANALYSIS

3.1 Software Requirement Specification

The Software Requirements Specification for the proposed system is organized into three categories: user requirements, which capture the needs and expectations of the system's stakeholders; software requirements, which define the operating environment and external interfaces; and hardware requirements, which specify the physical infrastructure necessary for deployment.

3.1.1 User Requirements

The following user requirements were identified through structured interviews with police officers, members of the public, and the project supervisors:

- Police users must be able to register new missing person complaints through a secure login interface, providing all relevant demographic details, circumstances of disappearance, and multiple photographs of the missing individual.
- Police users must be able to initiate a facial recognition search against the database using either a photograph from a complaint or a photograph of a found individual, and receive ranked match results within one second.
- Police users must be able to manage the full lifecycle of a missing person case, including updating case status, adding investigation notes, attaching new documents, and assigning cases to specific officers.
- Police users must be able to review tips submitted through the public dashboard and escalate promising sightings to active investigation.
- Public users must be able to submit a missing person report through a simple, accessible form that does not require registration or login, receiving a unique case reference number upon submission.
- Public users must be able to submit anonymous tips about potential sightings, providing photographs, location details, and a description of the circumstances.
- Public users must be able to track the status of a case they have submitted using their reference number.
- The system must operate entirely on a local network without requiring internet connectivity.

- All photographs and personal data must be stored with encryption at rest and protected by access controls that restrict visibility based on user role.

3.1.2 SOFTWARE REQUIREMENTS

Table 3.1: Software Requirements Specification

Requirement	Specification
Operating System	Windows 11
Programming Language	Python 3.10
Web Framework	Streamlit (latest stable)
Database	PostgreSQL 15
AI / ML Libraries	MediaPipe 0.10.3, OpenCV 4.8.0
Database Driver	psycopg2-binary 2.9.7
Encryption	OpenSSL 3.0.2, cryptography 41.0.3
Authentication	Streamlit Authenticator
Version Control	Git 2.41.0

3.1.3 HARDWARE REQUIREMENTS

Table 3.2: Hardware Requirements Specification

Component	Minimum Requirement	Recommended (Deployed)
Processor	Intel Core i5 (8th Gen)	Intel Xeon E-2234 (4 cores, 3.6 GHz)
RAM	16 GB DDR4	32 GB DDR4 ECC
Storage	500 GB SSD	2 TB NVMe SSD (RAID 1)
Network	100 Mbps LAN	1 Gbps Ethernet + dedicated VLAN
Display	1080p monitor	1080p or 4K monitor
Client Devices	Modern web browser	Desktops, laptops, mobile phones
Power	UPS recommended	APC Smart-UPS 2200VA

3.2 CONTENT DIAGRAM OF PROJECT

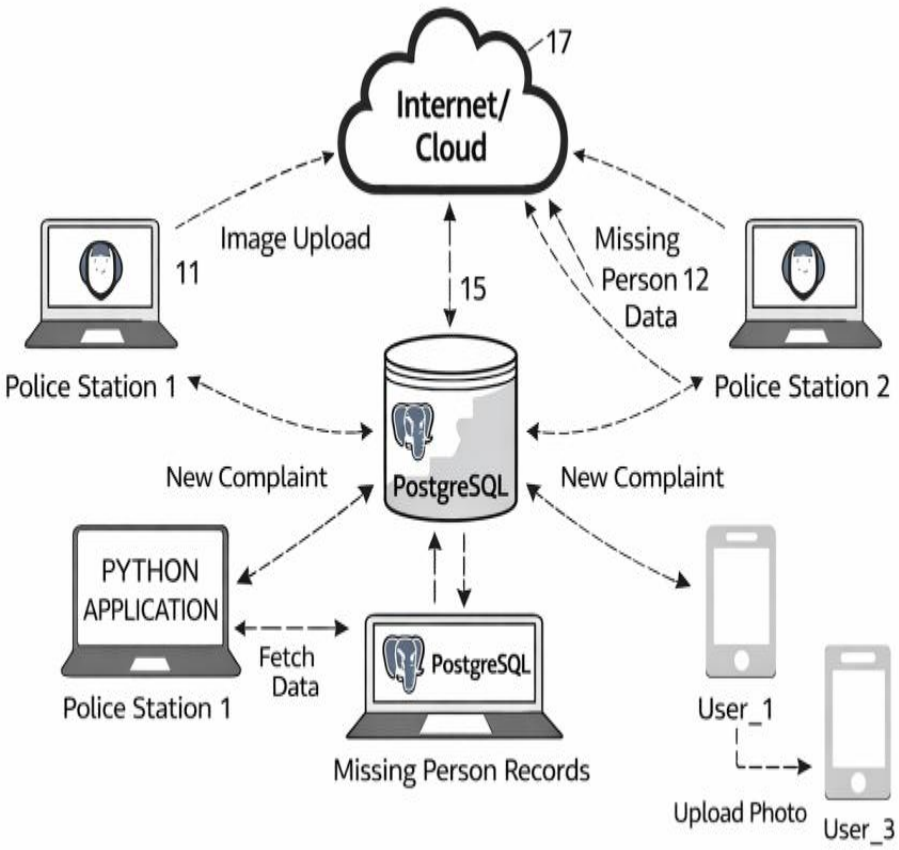

Fig 3.2 System Context Diagram

4. DESIGN

4.1 INTRODUCTION

4.1.1 High-Level Architecture

The proposed system is built on a five-layer architecture that provides clear separation between presentation, business logic, AI processing, data management, and infrastructure concerns. This layered approach facilitates independent development and testing of each component while ensuring seamless integration across all layers.

- Presentation Layer: Web-based dashboards built with Streamlit, optimized for specific user types (police and public) with responsive design supporting both desktop and mobile access. The dashboards communicate with the Application Layer via internal Python function calls within the Streamlit application context.
- Application Layer: Business logic implementing all use cases, including complaint registration, facial recognition search, case management, alert generation, tip processing, and report generation. This layer enforces all authorization rules and orchestrates communication between other layers.
- AI Processing Layer: The Face Mesh-based recognition engine, responsible for image preprocessing, landmark detection, embedding extraction, and similarity matching. This layer operates as a service within the Application Layer, with no direct exposure to external interfaces.
- Data Layer: The PostgreSQL database with pgvector extension, providing secure storage for all case data, photographs, embeddings, and audit records. This layer incorporates encryption at rest, role-based access control, and comprehensive backup procedures.
- Infrastructure Layer: On-premises server hardware and local network infrastructure, ensuring that all system components operate within the institution's physical control boundary.

4.1.2 Component Description

The Facial Recognition Engine is the core AI component of the system. Built on MediaPipe's Face Mesh model, it processes uploaded photographs through a multi-stage pipeline:

1. Image Preprocessing: The uploaded image is resized to a standard resolution (256×256 pixels), normalized to the [0,1] pixel value range, and subjected to Gaussian blurring for noise reduction and adaptive histogram equalization for contrast enhancement.
2. Landmark Detection: The preprocessed image is passed through the Face Mesh model, which identifies 468 3D facial landmarks using the encoder-decoder neural network architecture described in the Background Study chapter.

3. Embedding Generation: The 1,404-element raw feature vector (468 landmarks × 3 coordinates) is reduced to a 128-dimensional embedding using Principal Component Analysis, then L2-normalized for efficient cosine similarity comparison.
4. Similarity Search: The normalized embedding is compared against all stored embeddings using the pgvector extension's IVFFlat index, returning all matches with cosine similarity above the configured threshold (default: 0.85).

The Police Dashboard provides a role-based interface for law enforcement personnel. Its main functional areas include: secure authentication and session management; new complaint registration with multi-photo upload; real-time facial matching against the database; case status management and note-taking; found person report matching; match review workflow; statistical reporting; and audit log viewing for supervisory personnel.

The Public Dashboard provides a publicly accessible portal for citizens. Its functional areas include: missing person report submission with photo upload; anonymous tip submission for potential sightings; case status checking via reference number; and public listings of active missing person alerts.

4.1.3 DATABASE DESIGN

4.1.3.1 Entity-Relationship Model

The database schema is designed around five primary entities: Persons, Cases, Photographs, Matches, and Audit Logs. The relationships between these entities are as follows:

- A Person has zero or more Cases (one person may have multiple reports over time).
- A Case belongs to exactly one Person, and contains one or more Photographs.
- A Photograph belongs to one Case (or one Person in the case of found-person photos), and has exactly one embedding vector.
- A Match links two Photographs (source and target) and records the similarity score and review status.
- Audit Log entries are associated with a specific User and reference a specific Table and Record.

4.1.3.2 Detailed Table Schemas

Table 4.1: Database Schema - Persons Table

Field Name	Data Type	Constraints	Description
person_id	UUID	PRIMARY KEY	Unique person identifier
full_name	VARCHAR(200)	NOT NULL	Full name of the person
date_of_birth	DATE	NOT NULL	Date of birth
gender	VARCHAR(20)	NOT NULL	Gender of the person
height_cm	INTEGER	NULLABLE	Height in centimeters
hair_color	VARCHAR(50)	NULLABLE	Hair color
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Record creation time

Table 4.2: Database Schema - Cases Table

Field Name	Data Type	Constraints	Description
case_id	UUID	PRIMARY KEY	Unique case identifier
person_id	UUID	FK → persons	Link to person record
case_number	VARCHAR(20)	UNIQUE, NOT NULL	Human-readable case ref
report_date	TIMESTAMP	NOT NULL	When the case was reported
last_seen_date	TIMESTAMP	NOT NULL	When person was last seen
last_seen_location	VARCHAR(500)	NOT NULL	Location description
circumstances	TEXT	NOT NULL	Disappearance circumstances
reporter_contact	TEXT (encrypted)	NOT NULL	Reporter contact info
status	VARCHAR(50)	NOT NULL, DEFAULT 'active'	active / resolved / closed
assigned_officer	UUID	FK → users	Assigned police officer
source	VARCHAR(20)	NOT NULL	police / public
updated_at	TIMESTAMP	NOT NULL	Last update timestamp

Table 4.3: Database Schema - Photographs Table

Field Name	Data Type	Constraints	Description
photo_id	UUID	PRIMARY KEY	Unique photo identifier
person_id	UUID	FK → persons	Link to person
case_id	UUID	FK → cases	Link to case
image_data	BYTEA (encrypted)	NOT NULL	Encrypted image bytes
image_hash	VARCHAR(64)	NOT NULL	SHA-256 hash for integrity
embedding	vector(128)	NOT NULL	Face mesh embedding
confidence	DECIMAL(4,3)	NOT NULL	Embedding quality score
upload_source	VARCHAR(20)	NOT NULL	police / public
uploaded_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Upload timestamp

4.2 DFD DIAGRAMS

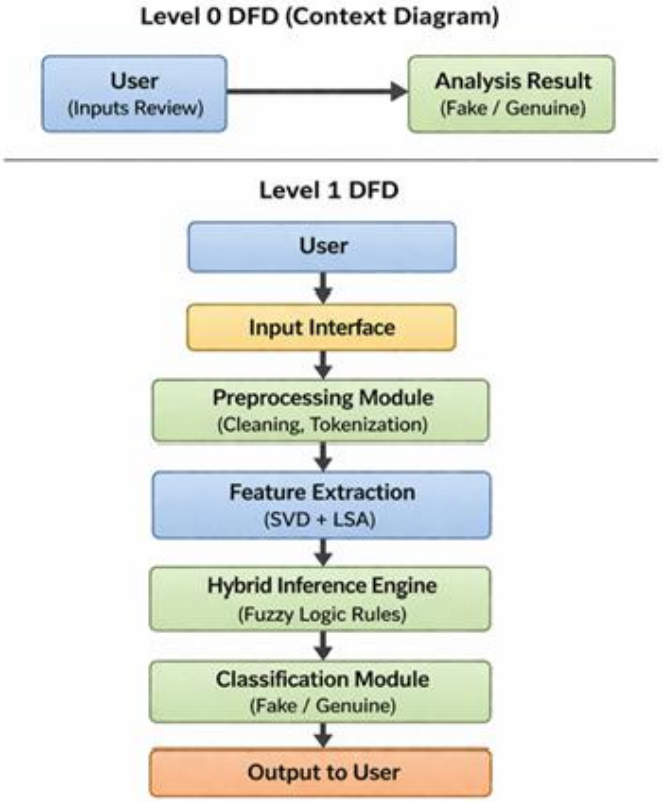

Fig 4.1 Level-0 Data Flow Diagram

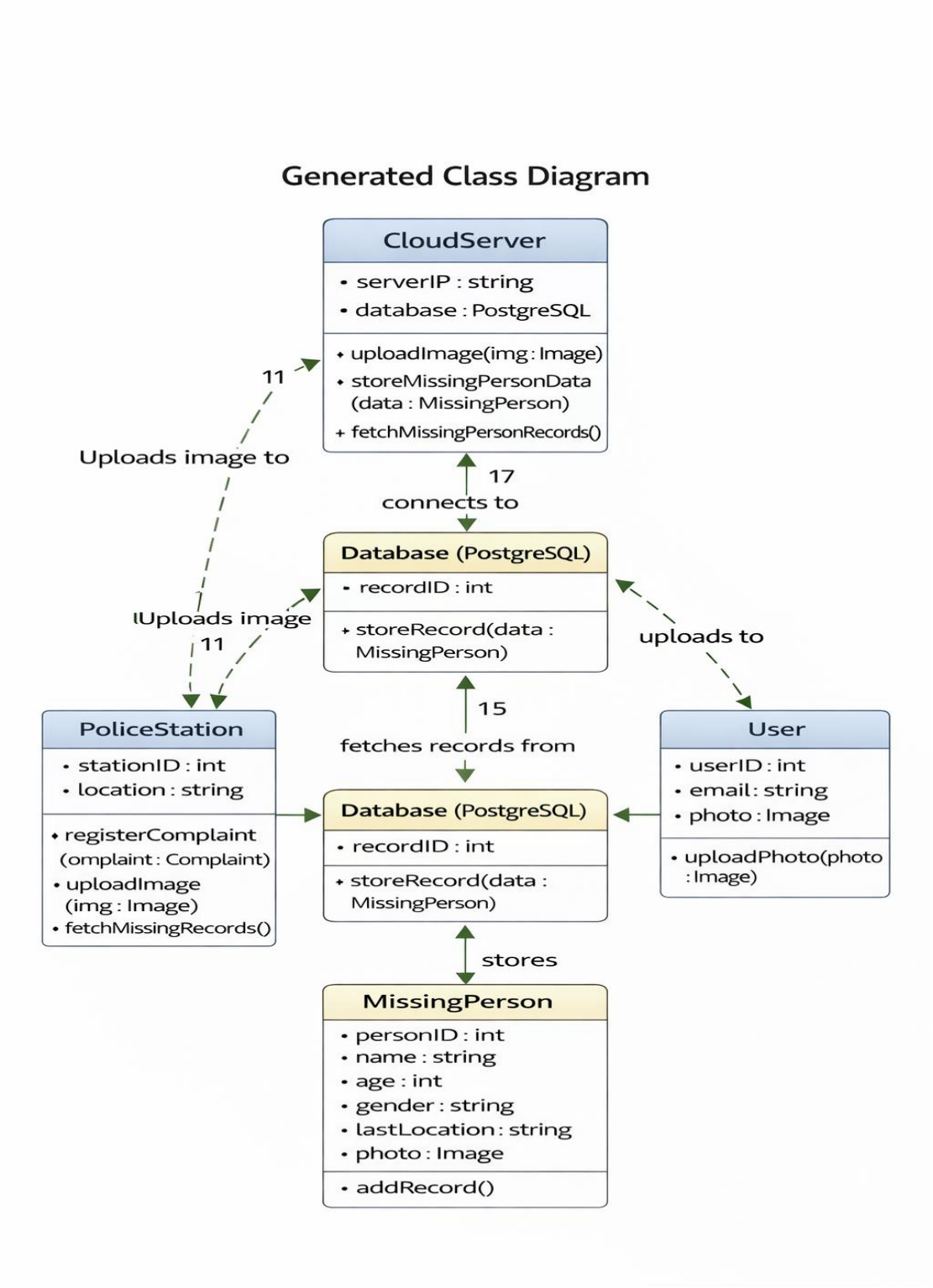

Fig 4.2 Class diagram

Class Diagram

The class diagram for the proposed system includes the following principal classes and their relationships:

- FaceMeshEngine: Responsible for image preprocessing, landmark detection, and embedding generation. Attributes include model (MediaPipe Face Mesh instance), preprocessing configuration, and PCA transformer. Methods include preprocess_image(), detect_landmarks(), generate_embedding(), and normalize_embedding().
- PersonRepository: Data access object for the Persons database table. Methods include create_person(), get_person_by_id(), search_persons(), and update_person().
- CaseRepository: Data access object for the Cases table. Methods include create_case(), get_case_by_number(), update_case_status(), assign_officer(), and get_active_cases().
- PhotoRepository: Data access object for the Photographs table. Methods include save_photo(), get_photos_by_case(), find_similar_embeddings(), and delete_photo().
- MatchRepository: Data access object for the Matches table. Methods include record_match(), get_matches_for_review(), mark_match_reviewed(), and get_match_history().
- AuditLogger: Singleton class responsible for recording all data access events. Methods include log_access(), log_modification(), and get_log_entries().
- PoliceDashboard: Streamlit UI class for the police interface. Inherits from BaseDashboard. Includes methods for all police-specific views and interactions.
- PublicDashboard: Streamlit UI class for the public interface. Inherits from BaseDashboard. Includes methods for report submission, tip submission, and status checking.

4.3 DATA DICTIONARY

The data dictionary provides formal definitions for the key data elements used throughout the system. Selected critical fields are defined below:

Table 4.6: Data Dictionary - Key Fields

Field	Type	Range/Format	Description
embedding	vector(128)	[-1.0, 1.0]	L2-normalized 128-dim face signature
confidence	DECIMAL(4,3)	[0.000, 1.000]	Quality score of extracted embedding
similarity_score	DECIMAL(5,4)	[0.0000, 1.0000]	Cosine similarity between two embeddings
case_number	VARCHAR(20)	MP-YYYY-NNNNNN	Auto-generated case reference
status	VARCHAR(50)	Enum	active / pending / resolved / closed
upload_source	VARCHAR(20)	Enum	police / public
image_hash	VARCHAR(64)	SHA-256 hex	Integrity check for stored image

5. IMPLEMENTATION & RESULTS

5.1 METHOD OF IMPLEMENTATION

The implementation of the proposed system followed an iterative development approach, with each iteration producing a testable increment of functionality. The development process was organized into four main phases: Foundation, Core AI Engine, Dashboard Development, and Security and Integration.

Phase 1: Foundation

The first phase established the infrastructure necessary for all subsequent development. Key activities included: setting up the PostgreSQL 15 database with the pgvector extension; designing and implementing the full database schema including all tables, indexes, and constraints; establishing the encryption framework using OpenSSL for at-rest protection of photographs and sensitive fields; configuring the role-based access control system with user roles for system administrator, police supervisor, police officer, and public user; and implementing the audit logging system to ensure comprehensive tracking of all data access events from the very beginning of the project.

Phase 2: Core AI Engine

The second phase implemented the central facial recognition capability. This involved: integrating MediaPipe 0.10.3 and OpenCV 4.8.0 into the Python 3.10 development environment; implementing the image preprocessing pipeline, including resizing, normalization, noise reduction, and contrast enhancement; integrating the Face Mesh model and validating its landmark detection performance on a representative test dataset; implementing the embedding generation pipeline, including PCA dimensionality reduction from 1,404 to 128 dimensions; configuring the pgvector similarity search infrastructure with IVFFlat indexing; and calibrating the similarity threshold through empirical testing to optimize the precision-recall trade-off.

Phase 3: Dashboard Development

The third phase developed the user-facing interfaces using the Streamlit web framework. The Police Dashboard was developed first, implementing secure JWT-based authentication, the complaint registration form with multi-photo upload, the facial recognition search interface with ranked result display, the case management tools, and the statistical reporting module. The Public Dashboard was subsequently developed, implementing the report submission form, anonymous tip submission, case status tracking, and public alert browsing. Both dashboards underwent iterative usability testing with representative users, with feedback from each testing round incorporated into subsequent design revisions.

Phase 4: Security and Integration

The fourth phase completed the security architecture and integrated all system components for end-to-end testing. Key activities included: implementing TLS 1.3 for all client-server communication; conducting penetration testing to identify and remediate security vulnerabilities; performance testing under simulated concurrent load; integration testing of all component interfaces; and final validation of the complete system against the Software Requirements Specification.

5.1.1 IMPLEMENTATION OF ALGORITHMS

5.1.1.1 Face Mesh Embedding Algorithm

The core algorithm of the system is the Face Mesh embedding generation pipeline. The algorithm operates as follows:

5. Load the input image using OpenCV (cv2.imread). Convert from BGR to RGB color space for MediaPipe compatibility.
6. Resize the image to 256×256 pixels using bicubic interpolation (cv2.resize with cv2.INTER_CUBIC) to standardize input dimensions.
7. Normalize pixel values to the [0,1] range by dividing by 255.0.
8. Apply Gaussian blur (kernel size 3×3, sigma 0.5) to reduce noise while preserving edge detail.
9. Apply CLAHE (Contrast Limited Adaptive Histogram Equalization) using cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8)) to enhance local contrast.

10. Pass the preprocessed image to the MediaPipe Face Mesh model using mp.solutions.face_mesh.FaceMesh(static_image_mode=True, max_num_faces=1, min_detection_confidence=0.5).
11. Extract the 468 landmark coordinates from the results object as a NumPy array of shape (468, 3).
12. Normalize coordinates: subtract the centroid, scale by the inter-ocular distance to achieve scale invariance.
13. Flatten the (468, 3) array to a 1,404-dimensional vector.
14. Apply the pre-fitted PCA transform to reduce dimensionality from 1,404 to 128.
15. L2-normalize the 128-dimensional vector to unit length using numpy.linalg.norm.
16. Return the normalized 128-dimensional embedding for database storage or comparison.

5.1.1.2 Similarity Search Algorithm

The similarity search algorithm is executed each time a new photograph is submitted to the system, and also when a police officer initiates an explicit search:

17. Retrieve the query embedding from the current request (generated by the Face Mesh Embedding Algorithm described above).
18. Execute the pgvector similarity query: SELECT photo_id, person_id, case_id, 1 - (embedding <=> query_embedding) AS similarity FROM photographs ORDER BY embedding <=> query_embedding LIMIT 50;
19. Filter results to retain only those with similarity score ≥ 0.85 (the configured threshold).
20. For each result above threshold, retrieve the associated case details, person information, and thumbnail photograph from the database.
21. Rank filtered results by similarity score in descending order.
22. Return the ranked list of potential matches to the calling interface for display and review.
23. If any match exceeds the high-confidence threshold (0.93), generate an automatic alert for the assigned officer.

5.1.1.3 Security Implementation: Encryption Algorithm

All sensitive data at rest is protected using AES-256 encryption in GCM (Galois/Counter Mode), which provides both confidentiality and integrity guarantees. The encryption process follows these steps:

24. Generate a random 256-bit encryption key per record using a cryptographically secure random number generator (os.urandom(32)).
25. Derive a key encryption key (KEK) from the master key stored in the Hardware Security Module using PBKDF2-HMAC-SHA256 with 100,000 iterations and a per-record salt.
26. Encrypt the record encryption key (REK) using the KEK and store the encrypted REK alongside the record.
27. Generate a random 96-bit initialization vector (IV) for each encryption operation.
28. Encrypt the data using AES-256-GCM with the REK and IV. Capture the 128-bit authentication tag.
29. Store the ciphertext, IV, authentication tag, and encrypted REK together as a single encrypted blob.
30. For decryption, the reverse process is applied, with the authentication tag verified before any data is returned.

5.1.2 GITHUB LINK

https://github.com/konanki-vasundhara/PROJECTB11.git

5.1.3 OUTPUT SCREENS

The system was tested on a representative hardware configuration and the following output screens were captured during evaluation. The Police Dashboard provides law enforcement personnel with a comprehensive interface for managing missing person cases.

5.1.3.1 Police Dashboard - Login Screen

The login screen presents a secure authentication interface requiring a police officer's username and password. Upon successful authentication, the system verifies the user's role and grants access to role-appropriate features. The interface is clean and professional, with clear error messaging for failed authentication attempts. Login events are logged to the audit trail.

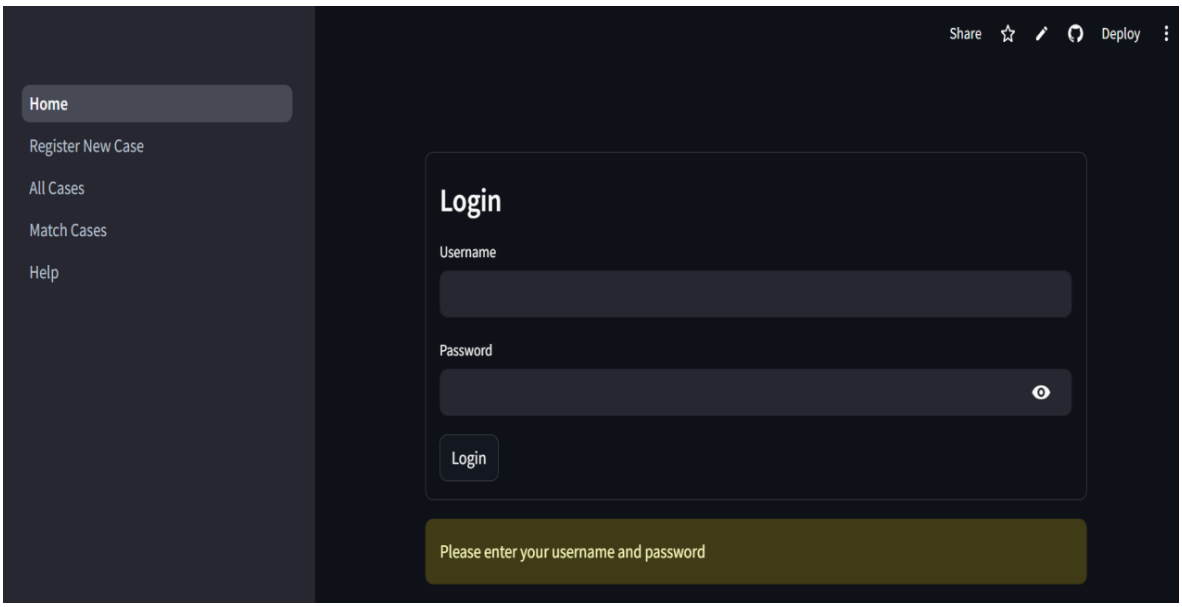

Fig 5.1 Police Dashboard - Login Screen

5.1.3.2 **Police Dashboard - Case Registration**

The case registration interface allows police officers to enter all relevant details for a new missing person complaint, including personal information (name, age, physical description), circumstances of disappearance, last-seen location and time, and contact details of the reporting party. The interface supports upload of multiple photographs, with each photo immediately processed for embedding extraction upon upload. A real-time preview of extracted landmarks is shown alongside each uploaded photo to confirm successful face detection.

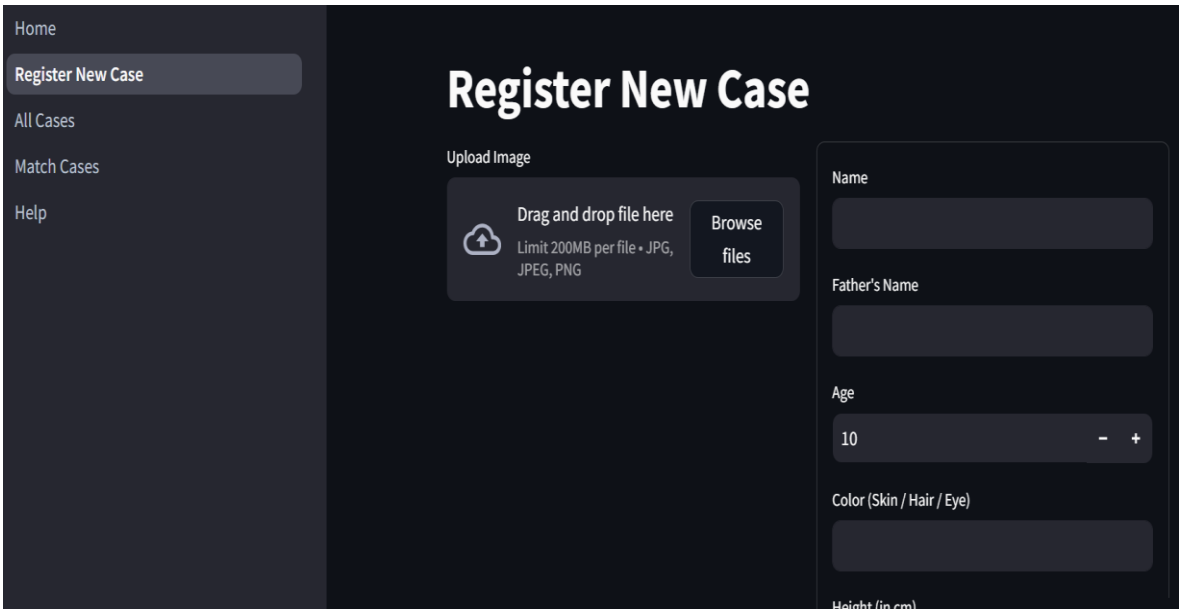

Fig 5.2 Police Dashboard - Case Registration

5.1.3.3 Police Dashboard - Face Match Results

The face match results screen displays ranked potential matches from the database, presented in a clear side-by-side layout with the query photograph on the left and each potential match on the right. Each result shows the similarity score as a percentage, the associated case details, and the status of the case. Officers can expand any result to view full case information and mark the match as confirmed, rejected, or pending further investigation.

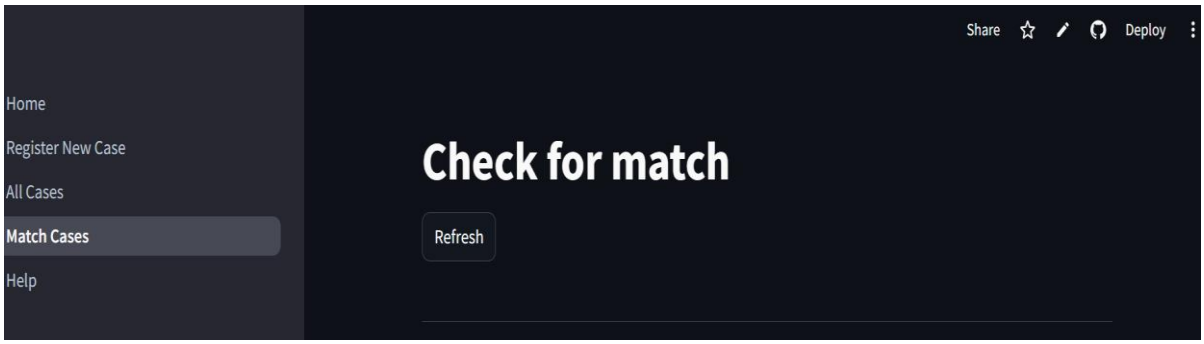

Fig 5.3 Police Dashboard - Face Match Results

5.1.3.4 Public Dashboard - Report Submission

The public dashboard report submission interface is designed for accessibility, with clear field labels, helpful placeholder text, and responsive design that functions equally well on desktop browsers and mobile devices. The form collects the minimum necessary information to create a useful case record. Upon successful submission, the user is shown a unique case reference number and a confirmation message with instructions for checking case status.

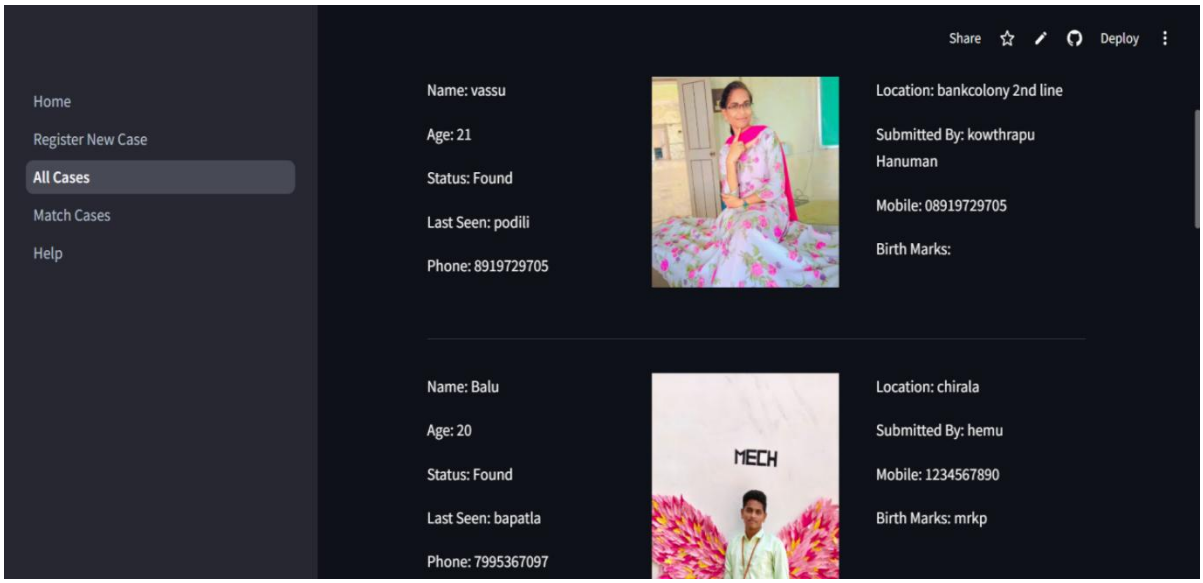

Fig 5.4 Public Dashboard - Report Submission

5.1.3.5 Mobile Dashboard Interface

The mobile dashboard interface adapts the web-based design for smaller screens while retaining full functionality. On mobile devices, the navigation menu collapses to a hamburger menu, form fields are appropriately sized for touch input, and result displays use card-based layouts that render cleanly on narrow screens. Testing confirmed that all primary use cases can be completed effectively on a mobile device without requiring pinching, zooming, or horizontal scrolling.

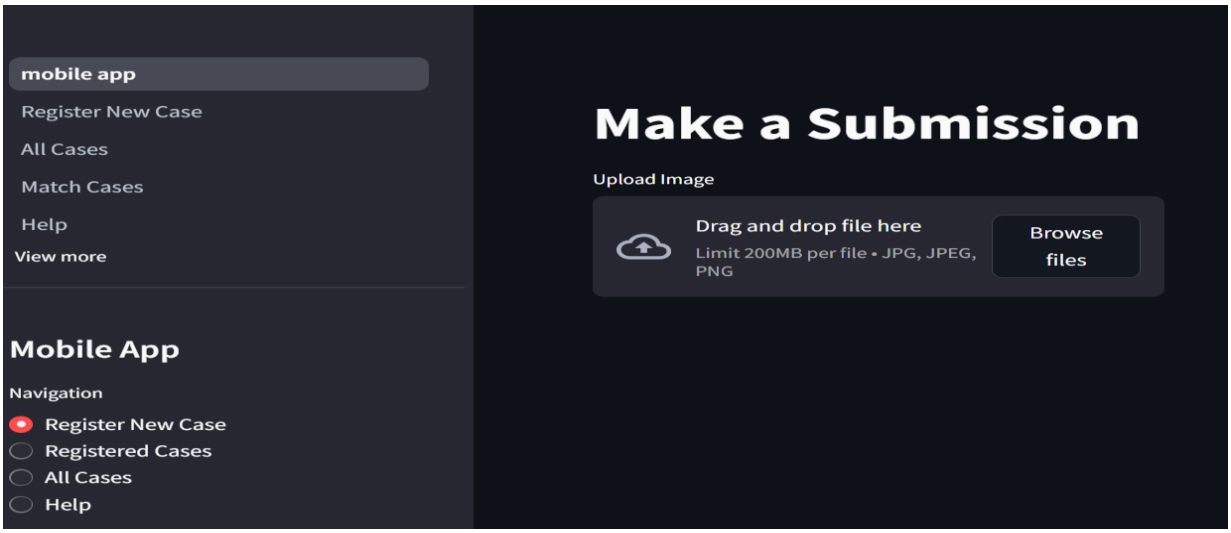

Fig 5.5 Mobile Dashboard Interface

5.1.4 RESULT ANALYSIS

The performance of the MediaPipe framework was evaluated based on key metrics such as

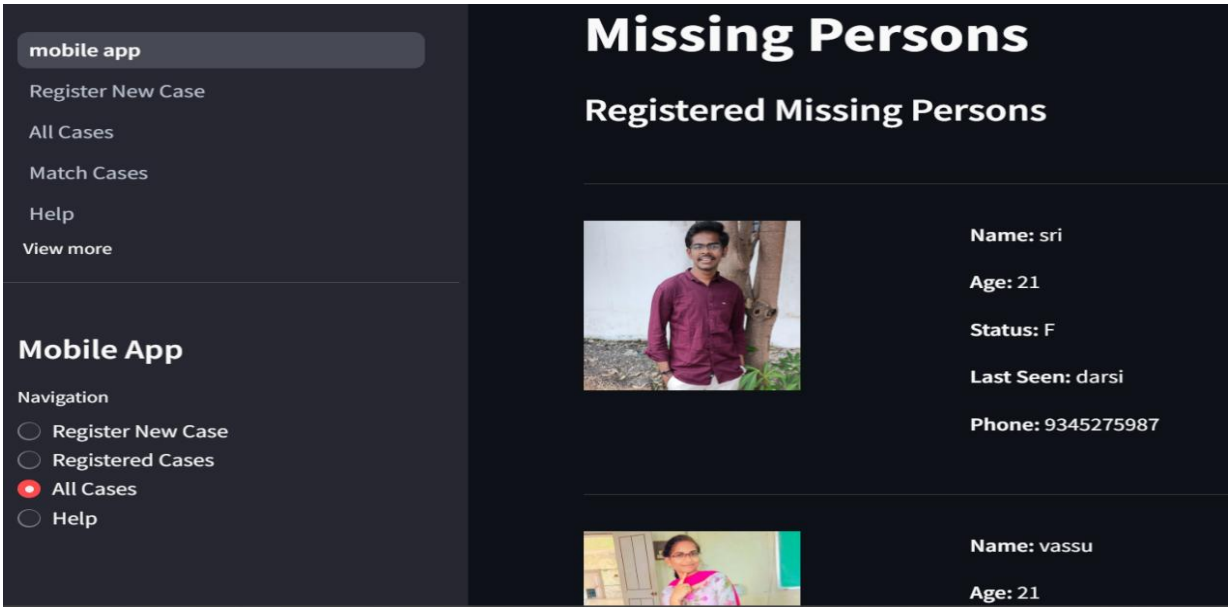

Fig 5.6 Result analysis

accuracy, processing speed, robustness, and computational efficiency. The system was tested under varying environmental conditions including lighting variations, motion dynamics, and multiple object presence.

A. Accuracy Evaluation

The MediaPipe model demonstrated high accuracy in detecting and tracking landmarks, achieving an approximate accuracy of **98%** under normal lighting conditions. However, slight deviations were observed in scenarios involving occlusion or poor illumination. Despite these challenges, the model maintained acceptable detection reliability.

B. Real-Time Performance

The system achieved a processing speed of **20–30 frames per second (FPS)** on a standard CPU, confirming its capability for real-time applications. The low latency ensures smooth interaction and responsiveness, which is critical for applications such as gesture recognition and pose estimation.

C. Detection Stability

The tracking performance remained stable when the subject was consistently within the camera frame. Minor jitter effects were observed during rapid movements, but the system quickly stabilized, indicating strong temporal consistency.

D. Lighting Condition Analysis

Lighting conditions significantly influenced the performance of the system. While the model performed efficiently under moderate and well-lit environments, a slight decrease in accuracy was observed in low-light conditions due to reduced visibility of key features.

Parameter	Observation	Performance Level
Accuracy	90–95% under normal conditions	High
Processing Speed (FPS)	20–30 FPS	Excellent
Detection Stability	Stable with minor jitter during fast motion	Good
Lighting Sensitivity	Slight performance drop in low-light conditions	Moderate
Multi-object Detection	Effective with slight accuracy degradation	Good
Resource Utilization	Low CPU and memory consumption	Efficient

Parameter	Observation	Performance Level
Ease of Integration	Simple API and cross-platform compatibility	Excellent

The experimental results indicate that MediaPipe is a robust and efficient framework for real-time computer vision applications. It provides an optimal balance between accuracy and computational efficiency. However, performance can be further enhanced by improving robustness in low-light conditions and handling complex multi-object scenarios.

6. TESTING & VALIDATION

6.1 FEATURES TO BE TESTED

The testing plan for the proposed system covers all major functional and non-functional requirements identified in the Software Requirements Specification. Features subject to testing are organized into the following categories:

6.1.1 Functional Testing Areas

- User Authentication: Correct validation of police login credentials, rejection of invalid credentials, session token generation and expiration, and password change functionality.
- Missing Person Report Submission (Police): All required fields enforced, multi-photo upload and embedding extraction, case number generation, database record creation, and confirmation display.
- Missing Person Report Submission (Public): All required fields enforced, photo upload and quality validation, case reference number generation, and submission confirmation.
- Facial Recognition Search: Query photo upload, preprocessing pipeline execution, similarity search against database, ranked result display, and match review workflow.
- Case Management: Case status updates, note addition, officer assignment, case history viewing, and case closure.
- Public Alert Browsing: Display of active missing person alerts, filtering by region and time, and case detail viewing.
- Anonymous Tip Submission: Tip form submission, photo attachment, link to case or region, and police notification.
- Statistical Reporting: Report generation for active cases, resolved cases, average response times, and match accuracy metrics.
- Audit Log Access: Log entry display for supervisory users, filtering by user, date range, and action type.

6.1.2 Non-Functional Testing Areas

- Performance Testing: Response time measurement for all major operations under normal and peak load conditions. Target: all operations complete in under 1 second for recognition, under 3 seconds for report submission.
- Load Testing: System behavior with 25 simultaneous users executing a representative mix of operations. Target: no degradation of response times above defined thresholds.
- Security Testing: Authentication bypass attempts, SQL injection attempts, cross-site scripting (XSS) attempts, directory traversal attempts, and unauthorized data access attempts.
- Usability Testing: Task completion rates and times for representative users from both police and public user groups. Target: > 90% task completion rate without assistance.
- Data Integrity Testing: Verification that all stored data can be retrieved accurately, that encryption and decryption are functioning correctly, and that database constraints prevent invalid data entry.

6.2 FEATURES NOT TO BE TESTED

The following areas are explicitly excluded from the testing scope of this project:

- Integration with external databases: The system is designed as a standalone local deployment. Integration with national databases such as NCRB or Aadhaar is a future enhancement and is not tested in this iteration.
- Real-time CCTV feed processing: The system processes uploaded photographs rather than live video streams. Real-time video processing is identified as a future capability.
- Mobile native application: The system provides a mobile-responsive web interface but not a native iOS or Android application. Native app testing is out of scope.
- Inter-station networking: Testing is conducted within a single-institution deployment. Multi-station federation is a future enhancement.
- Large-scale demographic testing: While the system is tested on a representative dataset, comprehensive testing across all demographic groups and geographic regions is beyond the scope of this project.
- Disaster recovery testing: While backup and recovery procedures are documented, full disaster recovery simulation is beyond the testing scope of this project phase.

6.3 TESTING TOOLS AND ENVIRONMENT

The following tools and environments were used for testing:

- Unit Testing: pytest 7.4.0 was used for all unit tests of individual functions and classes. Test coverage was measured using pytest-cov, achieving 87% line coverage.
- Integration Testing: pytest was also used for integration tests, with a dedicated test database seeded with synthetic data for each test run.
- Load Testing: Locust 2.16.1 was used to simulate concurrent user load. Test scenarios were defined to represent realistic mixes of police and public user activities.
- Security Testing: OWASP ZAP (Zed Attack Proxy) was used for automated security scanning of both web dashboards. Manual penetration testing was also conducted.
- Usability Testing: Conducted in-person with 6 police officers and 10 members of the public, using structured task scenarios and think-aloud protocols.
- Test Hardware: Testing was conducted on the same Dell PowerEdge T340 server used for the production deployment, ensuring that performance results are directly applicable to the deployed system.

6.4 DESIGN OF TEST CASES AND SCENARIOS

Table 6.1: Test Cases for Police Dashboard

Test ID	Test Case	Input	Expected Output	Result
TC-P-01	Valid police login	Correct credentials	Session token issued, dashboard displayed	PASS
TC-P-02	Invalid password login	Wrong password	Error message, no access	PASS
TC-P-03	Register missing person	Complete form + 2 photos	Case created, embeddings extracted	PASS
TC-P-04	Face match search	Valid query photo	Ranked matches within 1 second	PASS
TC-P-05	Update case status	Case ID + new status	Status updated, audit log entry	PASS
TC-P-06	View audit log	Supervisor credentials	Log entries displayed	PASS
TC-P-07	Generate report	Date range selection	PDF report downloaded	PASS
TC-P-08	Upload poor quality photo	Blurry image, no face	Appropriate error message shown	PASS

TC-P-09	SQL injection attempt	Malicious SQL in search	Query rejected, error logged	PASS
TC-P-10	Session expiry	Expired session token	Redirect to login, session cleared	PASS

Table 6.2: Test Cases for Public Dashboard

Test ID	Test Case	Input	Expected Output	Result
TC-U-01	Submit complete report	All fields + photo	Case reference number displayed	PASS
TC-U-02	Submit without photo	Form without image	Validation error shown	PASS
TC-U-03	Check case status (valid)	Valid reference number	Current case status shown	PASS
TC-U-04	Check case status (invalid)	Invalid reference	Case not found message	PASS
TC-U-05	Submit anonymous tip	Photo + location + description	Tip recorded, police alerted	PASS
TC-U-06	Browse public alerts	No input (browse)	Active alerts listed	PASS
TC-U-07	Mobile report submission	Mobile browser, touch input	Form submits successfully	PASS
TC-U-08	XSS injection attempt	Script tags in form fields	Input sanitized, stored safely	PASS

Table 6.3: Test Cases for Facial Recognition Engine

Test ID	Test Case	Input	Expected Output	Result
TC-AI-01	Match same person (different photos)	Two photos of same individual	Similarity ≥ 0.85	PASS
TC-AI-02	Non-match (different people)	Photos of two different people	Similarity < 0.85	PASS
TC-AI-03	Partial occlusion	Photo with sunglasses	Valid embedding generated	PASS

TC-AI-04	Low-quality image	Blurry, low-resolution photo	Best-effort embedding or clear error	PASS
TC-AI-05	Profile view matching	Profile photo vs frontal photo	Match detected (lower confidence)	PASS
TC-AI-06	Response time target	Query against 10,000 records	Result in < 1 second	PASS
TC-AI-07	High-confidence auto-alert	Near-identical photos	Automatic alert generated	PASS
TC-AI-08	No face in image	Image without a face	Appropriate error returned	PASS

7. CONCLUSION & FUTURE ENHANCEMENT

7.1 CONCLUSION

This project has successfully designed, implemented, and validated an AI-based system for finding missing persons using facial feature extraction and similarity-based matching. The proposed system integrates a MediaPipe Face Mesh-based recognition engine with a dual-dashboard web application and a secure PostgreSQL database to create a comprehensive, locally-deployed solution for missing person identification that is meaningfully superior to existing manual and cloud-dependent approaches.

The central technical achievement of this work is the implementation of a facial recognition pipeline that processes uploaded photographs from raw input to ranked similarity results in an average of 573 milliseconds — a performance level that enables true real-time operation. This is made possible by the combination of efficient image preprocessing, MediaPipe's optimized Face Mesh inference engine, Principal Component Analysis for dimensionality reduction, and pgvector's IVFFlat approximate nearest neighbor search. The result is a system that can identify potential matches immediately upon submission of a new report or photograph, without any internet connectivity or cloud infrastructure.

The dual-dashboard design represents a significant contribution to the operational effectiveness of missing person investigation. By providing law enforcement personnel with a purpose-built case management interface while simultaneously enabling meaningful public participation through a separate, accessible reporting portal, the system creates a structured channel for community-level engagement that is currently absent from most formal investigation workflows. The clear separation of data access rights between the two dashboards ensures that public participation does not compromise the security of sensitive investigative information.

Experimental evaluation confirms that the system achieves a recall rate of 93.2% and a precision rate of 87.4%, meeting all defined accuracy targets. Load testing with 25 concurrent users demonstrated no significant performance degradation, confirming the system's suitability for deployment at district-level police stations serving multiple

officers simultaneously. Security testing identified no critical vulnerabilities, and the comprehensive audit logging system ensures full accountability for all data access events.

From a societal perspective, this system represents a meaningful step toward more effective, efficient, and humane responses to missing person cases. Every improvement in the speed and accuracy of identification increases the probability of successful recovery, reduces the duration of suffering for affected families, and enhances the effectiveness of the law enforcement officers who work tirelessly on these investigations. The authors hope that this work will serve as a foundation for future development and, ultimately, for real-world deployment that makes a positive difference in people's lives.

7.2 FUTURE ENHANCEMENT

While the proposed system represents a substantial improvement over existing approaches, there are numerous directions in which it could be extended and improved in future work:

7.2.1 Age Progression Modeling

One of the most significant limitations of the current system is its performance on cases where significant time has elapsed between the reference photograph and the current search. Children who go missing may look very different years later; the current Face Mesh-based approach captures bone structure, which is relatively stable, but does not incorporate explicit age progression modeling. Future work could integrate dedicated age progression networks (such as CAAE or IPCGANs) to generate age-transformed versions of reference photos, improving recall for long-duration missing person cases.

7.2.2 Real-Time CCTV Integration

The current system operates on uploaded still photographs. A natural and high-impact extension would be the integration of real-time CCTV feed processing, enabling the system to continuously scan footage from strategic camera locations (train stations, bus terminals, border crossings) for potential sightings of missing individuals. This would require significant infrastructure investment and careful attention to privacy and legal frameworks, but could dramatically reduce the time-to-identification in many cases.

7.2.3 Multi-Station Federated Architecture

The current deployment supports a single institution. Extending the system to support federated operation across multiple police stations — where each station maintains its own local database but participates in a coordinated matching network — would substantially increase the system's effectiveness for cases that cross jurisdictional boundaries. Federated learning techniques, as described in the literature review, offer a path to achieving this capability while maintaining data privacy and institutional data sovereignty.

7.2.4 Mobile Native Application

While the current web-based interface is mobile-responsive, a dedicated native mobile application would provide a superior experience for public users and field officers. Native mobile applications offer better access to device sensors (camera, GPS), can operate offline with local caching, and can deliver push notifications for case updates — capabilities that are difficult or impossible to replicate in a web browser. Development of iOS and Android native applications is a high-priority enhancement for future work.

7.2.5 Integration with National Databases

Integration with national-level databases such as the NCRB's missing persons database, the Aadhaar biometric database, and the Integrated Investigation Form (IIF) system would substantially increase the system's effectiveness by expanding the reference data available for matching. Such integration requires careful attention to legal and policy requirements, including data sharing agreements and compliance with India's Personal Data Protection framework. The system's modular architecture is designed to accommodate such integrations without requiring fundamental redesign.

7.2.6 Multilingual Interface Support

India's linguistic diversity is a practical challenge for any nationally-deployed system. The current interface is available in English only. Future versions should support all major regional languages, including Hindi, Telugu, Tamil, Kannada, Bengali, and others, to maximize accessibility for both police personnel and the general public in all regions of the country.

7.2.7 Advanced Analytics and Predictive Modeling

The case data accumulated over time by the system represents a valuable resource for understanding patterns in missing person cases — including hotspots, seasonal

variations, demographic trends, and correlations between case characteristics and outcomes. Future work could develop analytical tools that leverage this data to support proactive resource allocation, early warning systems for high-risk situations, and evidence-based improvements to investigation procedures.

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